

rand algos practice

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1 1.7

suppose we choose a permutation π of the ordered set $N = \{1, 2, \dots, n\}$ uniformly at random from the space of all permutations of N . let $L(\pi)$ denote the length of the longest increasing subsequence in permutation π .

- for large n and some positive constant c , prove that $\mathbb{E}[L(\pi)] \geq c\sqrt{n}$.

thoughts:

there must be some symmetry thing here

i looked up hint that the product of lis and lds is n .

idk how to find the c .

sol.

the fact that $lis \cdot lds = n$ means that on average lis must be $\theta(\sqrt{n})$. this is because $\theta(lis) = \theta(lds)$ since they are symmetric.

- is the bound on (a) tight?

2 1.9

prove that FIND is $O(n)$

sol.

binning takes $O(n)$ to the input size.

each pivot being chosen has a chance of $\frac{1}{n}$. the sum of

$$\mathbb{E}[T(n)] = cn + \sum_1^{n-1} \Pr(n)T[n] = cn + \frac{1}{n}O(n^2) = O(n) \quad (1)$$